Grade 7 Mathematics

Strand 1: Number

Sub-strand 2: Number Operations

Lesson 1 — Mental Maths, Four Operations, Decimals, and Indices

Codes: B7.1.2.1.1–1.3, B7.1.2.2.1–2.3, B7.1.2.3.1–3.5

Teaching order inside this block (my recommendation): mental strategies & powers of 10 → written four operations → decimals in operations → **indices last** (indices are a new language for repeated multiplication; students need confident multiplication first).

1. What this lesson is about

Four skills in one block, in the order you need them:

1. **Mental maths** — fast tricks with powers of 10 and number properties.
2. **Written four operations** — add, subtract, multiply, divide carefully.
3. **Decimals in operations** — same skills, with the decimal point to respect.
4. **Indices (powers)** — a short way to write $2 \times 2 \times 2$.

Words to unpack first:

Word	Meaning
Power of 10	10, 100, 1 000, … (and 0.1, 0.01, …)
Mental maths	calculating in your head with strategies
Digit / decimal point	see Sub-strand 1
Repeated factors	the same number multiplied several times
Index (plural: indices)	the small number saying how many times to multiply (the *exponent*)
Base	the number being multiplied

2. Powers of 10 (B7.1.2.1.1)

Multiplying by 10, 100, 1 000 **shifts the decimal point to the right**. Dividing **shifts it to the left**.

Operation	Example	What moves
× 10	$3.4 \times 10 = 34$	one right
× 100	$3.4 \times 100 = 340$	two right
÷ 10	$34 \div 10 = 3.4$	one left
÷ 1 000	$3\,400 \div 1\,000 = 3.4$	three left
× 0.1	$3.4 \times 0.1 = 0.34$	one left (same as ÷ 10)
× 0.01	$3.4 \times 0.01 = 0.034$	two left (same as ÷ 100)

3. Mental strategies (B7.1.2.1.2–1.3)

Use **rounding** and **compensation**:

Example: 48×25

- $25 = (100)/(4)$, so $48 \times 25 = 48 \times (100)/(4) = 12 \times 100 = 1\,200$.

Example: $198 + 47$

- $200 + 47 = 247$, then subtract 2 → **245**.

Distributive property (useful mental shortcut):

$a(b + c) = ab + ac$

$7 \times 102 = 7(100 + 2) = 700 + 14 = 714$.

Word problem (B7.1.2.1.3): A pen costs GH■ 3. Buy 25. $25 \times 3 = 75$ → **GH■ 75**.

4. Written four operations (B7.1.2.2.1–2.2)

4.1 Add / subtract up to four digits

```
2 348
+ 1 756
------
4 104
```

Line up the place values. Borrow/carry carefully.

4.2 Multiply and divide

Multiplication: 234×12

```
234
× 12
-----
468 (2 × 234)
2340 (10 × 234)
-----
2808
```

Division: $8\,064 \div 24 = 336$ (use long division; 24 × 336 = 8 064).

5. Decimals in operations (B7.1.2.2.2–2.3)

Add / subtract: line up the **decimal points**.

```
124.3
0.275
+ 74.06
--------
198.635
```

Multiply decimals: multiply as whole numbers, then place the point.

- Total decimal places in the factors = decimal places in the answer.
- 0.07×0.02: $7 \times 2 = 14$; 2 + 2 = 4 places → **0.0014**.

Divide decimals: make the divisor whole first.

- $4.86 \div 0.3 = 48.6 \div 3 = 16.2$.

Story problems (B7.1.2.2.3): A shirt costs GH■ 47.50. Kofi buys 3 and pays with GH■ 200. $3 \times 47.50 = 142.50$; change = $200 - 142.50 = $ GH■ 57.50.

6. Indices — powers of numbers (B7.1.2.3)

6.1 What a^n means

a^n = \underbrace{a × a × ·s × a}_(n times)

2^3 = 2 × 2 × 2 = 8. a is the **base**, n is the **index** (exponent).

6.2 Repeated factors (B7.1.2.3.1–3.2)

24 = 2 × 2 × 2 × 3 = 2^3 × 3 (product of powers).

6.3 Zero as an index (B7.1.2.3.3)

Any non-zero number to the power 0 is **1**: 7^0 = 1, 100^0 = 1

6.4 Value of a number in index form (B7.1.2.3.4)

3^4 = 3 × 3 × 3 × 3 = 81.

6.5 Product of powers (B7.1.2.3.5)

a^m × a^n = a^(m+n)

2^3 × 2^4 = 2^(7) = 128.

Also useful: a^m ÷ a^n = a^(m-n), (a^m)^n = a^(mn)

7. Twenty fully worked examples

1. 3.4 × 10 = 34.
2. 3.4 × 100 = 340.
3. 34 ÷ 10 = 3.4.
4. 3.4 × 0.01 = 0.034.
5. 48 × 25 = 1 200 (via 100/4).
6. 198 + 47 = 245 (via 200 + 47 – 2).
7. 7 × 102 = 714 (distributive).
8. 2 348 + 1 756 = 4 104.
9. 234 × 12 = 2 808.
10. 8 064 ÷ 24 = 336.
11. 124.3 + 0.275 + 74.06 = 198.635.
12. 0.07 × 0.02 = 0.0014.
13. 4.86 ÷ 0.3 = 16.2.
14. 18.4 × 1.27 = 23.368 (to 2 d.p. → 23.37).
15. 2^3 = 8.
16. 2^3 × 2^4 = 2^7 = 128.
17. 7^0 = 1.
18. 3^4 = 81.
19. 5^6 ÷ 5^2 = 5^4 = 625.
20. Story: 3 shirts at GH■ 47.50, pay GH■ 200 → change GH■ 57.50.

8. Eight exercise sets × 5 questions

Exercise 1 — Powers of 10

1. 3.4 × 10

2. `3.4 × 100`
3. `34 ÷ 10`
4. `3.4 × 0.01`
5. **[Review]** Value of 6 in 9 463 285.

Exercise 2 — Mental strategies

1. `48 × 25`
2. `198 + 47`
3. `7 × 102`
4. `199 × 5`
5. **[Review]** Round 12.473 to 2 d.p.

Exercise 3 — Written addition & subtraction

1. `2 348 + 1 756`
2. `5 000 - 2 748`
3. `45 678 + 23 456`
4. `10 000 - 6 789`
5. **[Review]** Order 4 530 082; 4 350 820; 4 530 802.

Exercise 4 — Written multiplication & division

1. `234 × 12`
2. `8 064 ÷ 24`
3. `56 × 15`
4. `1 001 ÷ 7`
5. **[Review]** Round 4 738 to the nearest 100.

Exercise 5 — Decimal operations

1. `124.3 + 0.275 + 74.06` (to 1 d.p.)
2. `0.07 × 0.02`
3. `4.86 ÷ 0.3`
4. `18.4 × 1.27` (to 2 d.p.)
5. **[Review]** Round 0.006 845 to 2 s.f.

Exercise 6 — Story problems with decimals

1. 3 shirts at GH■ 47.50, pay GH■ 200. Find the change.
2. A pen costs GH■ 3. Cost of 25 pens.
3. 112 flies each 0.1 kg. Total mass.
4. `(0.07 × 0.02) ÷ 14`
5. **[Review]** Write 5 040 007 in words.

Exercise 7 — Indices

1. `2^3`
2. `2^3 × 2^4`
3. `7^0`
4. `3^4`
5. **[Review]** Value of 7 in 832 713.

Exercise 8 — Mixed / challenge

1. `5^6 ÷ 5^2`

2. `48 ÷ 6 + 5 × 2 - 3`
3. `234 × 12 - 56`
4. **[Review]** Round 84 730 to 2 s.f.
5. **[Review]** Order 999 999; 1 000 000; 999 099.

9. Full worked solutions

Solutions — Exercise 1

1. **34**. 2. **340**. 3. **3.4**. 4. **0.034**. 5. **60 000**.

Solutions — Exercise 2

1. `48 × 100 / 4 = 1 200`.
2. `200 + 47 - 2 = 245`.
3. `700 + 14 = 714`.
4. `200 × 5 - 5 = 995`.
5. **12.47**.

Solutions — Exercise 3

1. **4 104**. 2. **2 252**. 3. **69 134**. 4. **3 211**.
5. **4 350 820 < 4 530 082 < 4 530 802**.

Solutions — Exercise 4

1. **2 808**. 2. **336**. 3. **840**. 4. **143**. 5. **4 700**.

Solutions — Exercise 5

1. `198.635` → **198.6**. 2. **0.0014**. 3. **16.2**.
4. `23.368` → **23.37**. 5. **0.0068**.

Solutions — Exercise 6

1. `142.50`; change **GH■ 57.50**. 2. **GH■ 75**. 3. **11.2 kg**.
4. **0.0001**. 5. **five million, forty thousand, and seven**.

Solutions — Exercise 7

1. **8**. 2. `2^7` = **128**. 3. **1**. 4. **81**. 5. **700**.

Solutions — Exercise 8

1. `5^4` = **625**. 2. `8 + 10 - 3` = **15**. 3. **2 752**.
4. **85 000**. 5. **999 099 < 999 999 < 1 000 000**.

10. Cheat sheet

Skill	Rule
× / ÷ by powers of 10	shift the decimal point
Distributive	`a(b+c) = ab + ac`
Decimals add/sub	line up the point
Decimals multiply	add the decimal places

`a^n`	multiply a by itself n times
`a^0`	`= 1`
`a^m × a^n`	`= a^(m+n)`

NaCCA CCP B7.1.2 — plain English, British notation.